



BHABHA UNIVERSITY, BHOPAL (M.P.)

Scheme of Examination Programme – Bachelor of Technology Session 2021-22

Semester – VIII

Branch – Civil Engineering

			Periods per week Hours / week.				Maximum Marks (Theory Slot)			Marks for End Sem. Examination		
			I	T	P		End Sem. Exam.	Mid Sem Exam	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/ Assignment /Sessional	
1	CE-81	Advanced Structural Design-II (Steel)	3	1	2	5	70	20	10	30	20	150
2	CE-82	Geo-Technical Engineering -II	3	1	2	5	70	20	10	30	20	150
3	CE-83	Elective-II	2	1	-	3	70	20	10	-	-	100
4	CE-84	Elective-III	2	1	-	3	70	20	10	-	-	100
5	CE-85	Major Project	-	-	8	4	-	-	-	120	80	200
6	CE-86	Non-Destructive Testing (NDT)	-	-	4	2	-	-	-	-	50	50
7	CE-87	Group Discussion (Internal Assessment)	-	-	4	2	-	-	-	-	50	50
		Total	10	4	20	24	280	80	40	180	220	800

L: Lecture T: Tutorial P: Practical

One credit refers to one hour teaching in Theory, Tutorial and two hour in practical.

S. No.	Elective-II	Elective-III
	Subject Name	Subject Name
1	Pre-stress Concrete Design	Finite Element Method
2	Traffic Engineering	Waste disposal & Management
3	Urban & Town Planning	Geo-Informatics

Unit – I

Plate girder bridges (Riveted and welded)

Unit – II

Trussed girder bridges for railways and highways (IRC & IRS holding). Bearings for bridges.

Unit – III

Water Tanks: Pressed steel tanks, tanks with ordinary plates, square, rectangular, circular with hemispherical bottom and conical bottom.

Unit – IV

Chimneys: Guyed and self supporting steel stacks.

Unit – V

Bunkers, Silos & Tower

Reference Books: -

1. Design of Steel Structures – Ramammutham
2. Design of Steel Structures – Punia
3. Steel Str. by Ramchandra Vol II
4. Steel Str. by Arya & Ajmani
5. Design of steel structures – L.S. Negi

Unit – I

Shallow Foundations : Type of foundations shallow and deep. Bearing capacity of foundation on cohesion less and cohesive soils. General and local shear failures. Factors effecting B.C. Theories of bearing capacity - Prandle, Terzaghi, Balla, Skempton, Meyerh of and Hansan. I.S. code on B.c. Determination of bearing capacity. Limits of total and differential settlements. Plate load test.

Unit – II

Deep Foundation : Pile foundation, Types of piles, estimation of individual and group capacity of piles in cohesion less and cohesive soils. Static and dynamic formulae.. Pile load test, Settlement of pile group, Negative skin friction, under- reamed piles and their design. Piles under tension, inclined and lateral load Caissons. Well foundation. Equilibrium of wells. Analysis for stability tilts and shifts, remedial measures.

Unit – III

Soil Improvement Techniques : Compaction. Field and laboratory methods, Proctor compaction tests, Factors affecting compaction. Properties of soil affected by compaction. Various equipment for field compaction and their suitability. Field compaction control. Lift thickness. Soil stabilisation : Mechanical, Lime, Cement, Bitumen, Chemical, Thermal, Electrical- stabilisation and sabilisation by grouting. Geo-synthetics, types, functions, materials and uses.

Unit – IV

Soil Exploration and Foundations on Expansive and Collapsible soils : Methods of soil exploration. Planning of exploration programme for buildings, highways and earth dams. Disturbed and undisturbed samples and samplers for collecting them. Characteristics of expansive and collapsible soils, their treatment, Construction techniques on expansive and collapsible soils. CNS layer.

Unit – V

Sheet piles/Bulkheads and Machine foundation : Classification of sheet piles/bulkheads.

Cantilever and anchored sheet piles, Cofferdams, materials, types and applications.

Modes of vibration. Mass-spring analogy, Natural frequency. Effect of vibration on soils.

Vibration isolation. Criteria for design. Design of block foundation for impact type of machine.

Reference Books: -

1. Soil Mechanics & Foundation Engg. by Dr. K.R. Arora - Std. Publishers Delhi
2. Soil Mechanics & Foundation Engg. by B.C. Punmia - Laxmi Publiscations Delhi
3. Modern Geotech. Engg. by Dr. Alam Singh-IBT Publishers Delhi.
4. Geotech. Engg. by C.Venkatramaiah-New AGE International Publishers, Delhi
5. Found. Engg. by GALEonards McGraw Hill Book Co.Inc.
6. Relevant IS Code

LIST OF EXPERIMENTS:

1. Indian Standard Light Compaction Test/Std. Proctor Test
2. Indian Standard Heavy Compaction Test/Modified Proctor Test
3. Determination of field density by Core Cutter Method
4. Determination of field density by Sand Replacement Method
5. Determination of field density by Water Displacement Method
6. The corifiled Compression Test
7. Triaxial compression test
8. Lab. Vane Shear test
9. CBR Test
10. Demonstration of Plate Load Test SPT & DCPT

BHABHA UNIVERSITY, BHOPAL

Branch- Civil Engineering

Subject – Elective-II(Pre-Stress Concrete Design)

Sub Code – CE 83-(I)

UNIT-I

Introduction– Theory And Behaviour: Basic concepts, Advantages, Materials required, Systems and methods of pre stressing–Analysis of sections, Stress concept, Strength concept, Load balancing concept, Effect of loading on the tensile stresses in tendons, Effect of tendon profile on deflections, Losses of pre-stress.

UNIT-II

Design Concepts: Flexural strength, Simplified procedures as per codes, strain compatibility method, Basic concepts in selection of cross section for bending– stress distribution in end block, Design of anchorage zone reinforcement, Limit state design criteria, Partial pre-stressing Applications.

UNIT III

Pre-stressed Concrete Slabs: One way slab two way slabs, pre-stressed concrete beam slab construction, pre-stressed flat slab, Deflection and Crack Width : Factors influencing deflection, short term deflections of un cracked members, long term deflection, deflections of cracked members.

UNIT IV

Miscellaneous Structural Members : Columns subjected to combined bending and axial force, piles, poles, piers and abutments and Tension member–ring beams. Design of pre- stressed circular concrete tanks –Pipes

UNIT V

Pre-Stressed Concrete Bridges: General aspects–pre-tensioned pre-stressed bridge decks

– Post tensioned post-stressed bridge decks – Principles of design only

Reference Books: -

1. Krishna Raju N., Prestressed concrete, Tata McGraw Hill Company, New Delhi 1998
2. Mallic S.K. and Gupta A.P., Prestressed concrete, Oxford and IBH publishing Co. Pvt. Ltd. 1997.
3. Rajagopalan, N, "Prestressed Concrete", Alpha Science, 2002
4. Jain & Jai Krishna, Plain & Reinforced Concrete Vol-II Nem chand & Bros Roorkee.
5. P. Dayaratran, Pre-stressed Concrete Structures, Oxford & IBH Co. Delhi
6. Ramaswamy G.S., Modern prestressed concrete design, Arnold Heinimen, New Delhi, 1990
7. Lin T.Y. Design of prestressed concrete structures, Asia Publishing House, Bombay 1995.
8. David A. Sheppard, William R. and Philips, Plant Cast precast and prestressed concrete
9. IS 1343-980 code of Practice for Pre-stressed Concrete < Bureau of India Standards New Delhi

Unit -I.

Traffic Characteristics : Road user's characteristics - general human characteristics, physical, mental and emotional factors, factors affecting reaction time, PIEV theory. Vehicular characteristics: Characteristics affecting road design-width, height, length and other dimensions. weight, power, speed and braking capacity of a vehicle.

Unit -II.

Traffic Studies : Spot Speed Studies and Volume Studies. Speed and Delay Studies purpose, causes of delay, methods of conducting speed and delay studies. Origin and destination Studies (O & D) : Various methods, collection and interpretation of data, planning and sampling. Traffic Capacity Studies: Volume, density, basic practical and possible capacities, level of service. Parking Studies: Methods of parking studies cordon counts, space inventories, parking practices.

Unit -III.

Traffic Operations And Control : Traffic regulations and various means of control. One way streets-advantages and limitations. Traffic signals- isolated signals, coordinated signals, simultaneous, alternate, flexible and progressive signal systems. Types of traffic signals, fixed time signals, traffic actuated signals, speed control signals, pedestrian signals, flashing signals, clearance interval and problems on single isolated traffic signal.

Unit -IV.

Street Lighting : Methods of light distribution. Design of street lighting system. Definitions- Luminaire, foot candle, Lumen, utilization and maintenance factors. Different types of light sources used for street lighting. Fundamental factors of night vision.

Unit -V.

Accident Studies & Mass Transportation : Accident Studies : Causes of accidents, accident studies and records, condition and collision diagram, preventive measures. Expressways and freeways, problems on mass transportation and remedial measures, brief study of mass transportation available in the country.

Reference Books :-

1. Traffic Engineering and Transport Planning by L.R. Kadiyali, Khanna Publishers, Delhi
2. Traffic Engineering by Matson, W.S. Smith & F.W. Hurd
3. G.J. Pingnataro, Principles of Traffic Engineering
4. D.R. Drew, Traffic Flow Theory
5. W.R. McShane and R.P. Roess "Traffic Engg"
6. Wohl & Martin, Traffic System Analysis for Engineering & Planner

BHABHA UNIVERSITY, BHOPAL

Branch- Civil Engineering

Subject – (Elective-II) Urban & Town Planning

Sub Code – CE 83-(III)

UNIT-I

Definition and classification of urban areas - Trend of urbanization - Planning process - Various stages of the planning process - Surveys in planning. Plans - Delineation of planning areas. utility of spaces, future growth etc. Role of “Urban Planner ”in planning and designing in relation with spatial organization, utility, demand of the area and supply

UNIT-II

Plan implementation- Urban Planning agencies and their functions - Financing- Public, private, Nongovernmental organizations- Public participation in Planning. Development control regulations. sustainability and rationality in planning, Components of sustainable urban and regional development, Emerging Concepts: Global City, inclusive city, Safe city, etc. City of the future, future of the city.

UNIT-III

Town and country planning act- Building bye-laws. Elements of City Planning, Zoning and land use, Housing. Introduction to landscaping, importance , objectives, principles, elements, Urban Planning standards Urban renewal for quality of life and livability.

UNIT-IV

Traffic transportation systems: urban road, hierarchy, traffic management, Intelligent Transport Systems. Legal Issues in Planning and Professional Practice, Concepts and contents related to planning provision regarding property rights, Concept of Arbitration, State and Central government to deal with various matters concerning Town and Country Planning. mechanism for preparation of DP: Land Acquisition Rehabilitation and Resettlement Act 2013.

UNIT-V

Types of Development plans: Master Plan, City Development Plan, Structure Plan, housing, land use, Water Supply & sanitation, etc., Planning agencies for various levels of planning. Their organization and purpose (CIDCO-MHADA-MIDC, MMRDA/ PMRDA etc),.

References Books-

1. Urban Planning: Theory & Practice By M. Pratap Rao
2. Urban Transportation: Planning, Operation and Management by S. Ponnuswamy, D. Johnson Victor
3. Sustainable Urban Planning in India by Joy Sen
4. Urban Planning in India by Amiya Kumar Das

BHABHA UNIVERSITY, BHOPAL

Branch- Civil Engineering

Subject – Elective-III(Finite Element Method)

Sub Code – CE 84-(I)

UNIT-I:

Introduction to Finite element method: General applicability and description of finite element method, comparison with other methods.

UNIT-II

:Solution of finite element method: Solution of equilibrium problems, eigen value problems, propagation problems, computer implementation of Gaussian eliminations, Choleskis decomposition, Jacobis and Ranga-Kutta method.

UNIT-III:

General procedure of finite element method: Descretization of the domain, selection of shapes, types and number of elements, node numbering technique, interpolation, polynomials, their selection and derivation in terms of global and local coordinates, convergence requirements. Formulation of element characteristic matrices and vectors, variational approach.

UNIT-IV:

Iso-parametric formulation: Lagrange and Hermite interpolation functions, iso- parametric elements, numerical integration.

UNIT-V

Static analysis: Formulation of equilibrium equation, analysis of truss, frames, plane stress and plane strain problems.

References Books-

1. Weaver, Johnson, Finite element and structural analysis
2. HC Martin, Matrix structural analysis
3. CF Abel, CS Desai, Finite element methods
4. Buchanan, Finite element Analysis (Schaum Outline S), TMH
5. Krishnamurthy, Finite element analysis, TM

BHABHA UNIVERSITY, BHOPAL

Branch- Civil Engineering

Subject –Elective-III(Water Disposal & Management)

Sub Code CE 84-(II)

UNIT – I

Solid Waste And Its Management: Introduction to solid waste-Definition of solid waste, garbage, rubbish-Sources and Types. Characteristics of Solid Wastes: Physical, chemical and biological characteristics. 3Rs- Reduction, reuse, recycling and recovery principles of waste management- Functional elements of Solid Waste management- Waste generation and handling at source-Collection of solid wastes- Collection methods and services- guidelines for collection route layout.

UNIT- II

Waste Water And Its Management: Characteristics and analysis of waste water, recycles of decomposition, physical, chemical & biological parameters. Unit operations for waste water treatment, preliminary treatment.

UNIT- III

Hazardous Waste And Its Management: Introduction to hazardous waste management issues, classification; Magnitude of problem; Risk assessment; Environmental Legislation; Characterization and site assessment; Waste minimization and resource recovery; Storage and Transportation of Hazardous wastes; Hazard in processing and treatment; Physical, Chemical, Thermal and Biological processes;

UNIT -IV

Transfer And Transport Of Wastes: Solid Waste transportation: Transfer station- Processing and segregation of the solid waste- various methods of material segregation. Hazardous waste transportation: According to economic benefits, transfer stations, and types of hazardous waste.

UNIT -V

Disposal Of Wastes: Solid waste disposal: Volume reduction, Open dumping, land filling techniques. Landfills: classification-Design and Operation of landfills, Land Farming, Deep well injection. Hazardous waste disposal; Landfill disposal and land storage; Ground water contamination; Containment; remedial alternatives. Natural methods of waste water disposal i.e. by land treatment & by dilution, self purification capacity of stream, Oxygen sag analysis.

References Books-

1. Waste Water Engg. by B.C. Punmia - Laxmi Publication (P) Ltd. New Delhi
2. Environmental Engg. - M.L. Davis & D.A. Cornwell - Mc Graw Hill Company
3. Integrated Solid Waste Management by Tchobanogous
4. Environmental Engineering by Howard S. Peavy, Donald R. Rowe and George.
5. Hazardous Waste Management by Charles A. Wentz - - Mc Graw Hill Company
6. Hazardous Waste Management [Gaynor W. Dawson](#), [Basil W. Mercer](#)

UNIT I

Basic Concepts Of Gis Basics of Geographic Information System (GIS): , Definition, Evolution & Components. GIS data input devices like scanner, digitizer, GPS, Remote sensing etc. Manual and semi-automatic line following digitization. Data Model: Raster Data Model, Grid , Tessellations. Spatial and attribute data visualization and query. Open GIS consortium, Customization in GIS , Object Oriented GIS, Web GIS, Introduction to popular GIS software like Arc GIS and QGIS.

UNIT II

Geo Informatics In Urban Mapping And Management: Remote sensing for detection of urban features. Introduction & basic terminology. Digital image processing techniques – Case studies. Segmentation of Built-up areas – Classification algorithms – Land use/ Land cover mapping – change detection – high resolution remote sensing – case studies. Mapping transportation network –Alignment planning – Traffic and parking studies – Accident analysis – case studies. Urban growth modeling – Expert systems in planning.

UNIT III

Basic Concepts Of Photogrammetry :History and development:- Types of aerial photo, Classification of aerial cameras, Scale, Overlaps, Stereoscopy, Concepts, Viewing and measuring systems, Image and object coordinates, floating mark, parallax equation, height information, Tilt , Rectification , Displacement. Flight planning, computation for flight plan, photo control, cost estimation, aerial mosaics, types. Concepts of interior, relative, absolute orientation, object, image relation, linearization, effect of orientation elements , scaling and leveling , analytical procedures , map compilation using stereo plotters.

Introduction to digital photogrammetry, Elements of Aero triangulation and analytical method, strip and block adjustment, Terrestrial photogrammetry.

UNIT IV

GPS Surveying: Introduction & components of GPS, Space segment, control segment and user segment, Elements of Satellite based surveys-Map datums, GPS receivers, GPS observation methods and their advantages over conventional methods.

UNIT V

Remote Sensing: Principle, components, classification, application on environmental engineering, hydrology and water resources , agriculture & forestry, disaster management.

Microwave Remote Sensing:Introduction, basic concepts, terminology and sensors in MWRS. Radar basics, radar interaction with earth surface ,geometry of radar images, radar return and image signature, resolution concepts.

References Books-

- 1.Plane Surveying- A. M. Chandra, New Age International.
- 2.Surveying and Leveling-Part-I & IIT- P. Kanetkar and S. V. Kulkarni, Vidyarthi Griha Prakashan.
- 3.Geoinformation: Remote Sensing, Photogrammetry and Geographic Information Systems- Gottfried Konecny, CRC Press
- 4.Elements of Photogrammetry- Paul R.Wolf, McGraw-Hill
- 5.Photogrammetry, Vol 1&II - Karl Kraus, Walter de Gruyter
- 6.Remote Sensing and image interpretation- Lillesand T.M. and Kiefer R. W., Wiley.
- 7.Introduction to remote sensing - J. B. Campbell, John Wiley.
- 8.Introductory digital image processing- J. R., Jensen Prentice Hall.
9. Remote Sensing in Civil Engineering- Kennie, T. J. M. and Matthews M. C., Surrey University Press.
10. GPS satellite surveying- Alfred Leick,.Wiley
11. GPS Theory, Algorithms and Applications- Guocheng Xu, Springer
12. Microwave remote sensing vol-1,vol-2- Ulaby,F.T.,Moore,K.R. and Fung,Artech House Publishers.
13. Principles and applications of Imaging - Floyd. M. Handerson Anthony, J.Lewis, Wiley.
14. Air and space borne radar systems-An introduction- Philippe Lacomme and Eric Normant, Elsevier.
15. Introduction to microwave remote sensing- Iain H.woodhouse,CRC Press
16. Satellite Remote Sensing for Hydrology and Water Management- Eric C. Barrett, Clare H.Power, Taylor & Francis Ltd
17. Hydrologic and Hydraulic Modeling Supportwith Geographic Information Systems-Dr. David Maidment, Dr. Dean Djokic, Esri Press.

BHABHA UNIVERSITY, BHOPAL

Branch- Civil Engineering

Subject –Non-Destructive Testing(NDT)

Sub Code – CE 86

List of experiment:

1. To study of Rebound Hammer Test.
2. To study of UPV Test.